One commenter observed another gap in information between the exclusive SIPs and proprietary market data. This commenter observed that when market-wide circuit breakers tripped, proprietary feeds continued to disseminate information, such as information on quotes, during the halt while the exclusive SIPs provided updates that were not in real-time.[1684]

Currently, the gap in information between data in the exclusive SIP and proprietary DOB products may limit the current level of price efficiency if market participants with access to proprietary DOB products do not incorporate this information into prices observed by exclusive SIP subscribers quickly enough through their trading or quoting activity.[1685] However, the Commission does not know the extent of this possible effect because it does not know how quickly market participants that subscribe to proprietary DOB products incorporate the information contained in these feeds into the information contained in the exclusive SIP.

(b) Current Process for Dissemination of SIP Data and Proprietary Data Feeds

Today, SIP data is disseminated to investors and market participants through a centralized consolidation model with an exclusive SIP for each NMS stock, centrally collecting market data transmitted from the dispersed SRO data centers and then redistributing the consolidated market data to market participants who are often in different locations.[1686] The SROs typically transmit their market data through fiber optic cables to the SIPs.[1687]

1684 See T. Rowe Price Letter at 2.

1685 For example, price efficiency may be limited if there is a delay in incorporating imbalance information observed in proprietary DOB feeds into the quote and trade prices shown by the exclusive SIP. See infra Section V.D.1. Price efficiency is greater when prices reflect current information faster.

1686 See Proposing Release, 85 FR at Sections I, II.A; see also DOJ Letter at 2.

1687 See Proposing Release, 85 FR at Section II.A.

Typically, proprietary data is transmitted directly from each exchange to the data center of the subscriber and does not first travel to a centralized consolidation location. Furthermore, unlike the standardized transmission of SIP data over fiber optic cable, proprietary data is frequently transmitted using low-latency wireless connectivity (e.g., microwave signals) or other forms of connectivity (often provided by the exchanges) that are faster than fiber.[1688] As stated by one commenter, data transmission via microwave signals is much faster than via fiber optic cables, because "microwave signals travel at the speed of light through air, rather than over fiber, which can attenuate signals."[1689]

There is a significant latency differential between SIP data and the proprietary market data products that are delivered directly to market participants or to market data aggregators who generally have better connectivity, communications, and aggregation technology than the SIPs.[1690] Specifically, the centralized consolidation model has three sources of latency: (a) geographic latency; (b) aggregation or consolidation latency; and (c) transmission or communication latency. The latency differentials between SIP data and proprietary data, are

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1688 Id.

1689 See Data Boiler Letter I at 39.

1690 See supra note 397; see also Robert P. Bartlett, III and Justin McCrary, *How Rigged Are Stock Markets? Evidence from Microsecond Timestamps* at 45 (2017), available at https://www.law.berkeley.edu/wp-content/uploads/2019/10/bartlett_mccrary_latency2017.pdf (''[O]ur analysis suggests SIP reporting latencies generate remarkably little scope for exploiting the informational asymmetries available to subscribers to exchanges' direct data fees.'').

meaningful, and market participants believe these differentials impact their ability to trade and their order execution quality.[1691]

Geographic latency refers to the time it takes for data to travel from one physical location to another. Greater distances usually equate to greater geographic latency, though geographic latency is also affected by the mode of data transmission. The Commission understands that geographic latency is typically the most significant component of the additional latency that SIP data feeds experience compared to proprietary data feeds.[1692] The record in this rulemaking suggests that the geographic latency of SIP data may be up to a millisecond.[1693]

1691 See Proposing Release, 85 FR at n.412 and accompanying text; Martin Scholtus et al., *Speed, Algorithmic Trading, and Market Quality around Macroeconomic News Announcements*, 38 J. BANKING & FIN. 89 (2014) ("This paper documents that speed is crucially important for high-frequency trading strategies based on U.S. macroeconomic news releases. Using order-level data on the highly liquid S&P 500 ETF traded on Nasdaq from January 6, 2009, to December 12, 2011, we find that a delay of 300 ms or more significantly reduces returns of news-based trading strategies."); Grace Hu et al., *Early peek advantage? Efficient price discovery with tiered information disclosure*, 126 J. FIN. ECON. 399 (2017) ("Calibrating the speed of price discovery at a finer scale, we find that the first 200 milliseconds at 9:54:58 accounts for 89% of the one-second return at 9:54:58 on negative news days, and 85% of the one-second return at 9:54:58 on positives news days. In other words, most of the price discovery happens during the first 200 milliseconds, faster than the blink of an eye."); Tarun Chordia et al., *Low Latency Trading on Macroeconomic Announcements,* 31 REV. FIN. STUD. 4650 (2018) ("Specifically, trading in the direction of the announcement surprise results in average dollar profits (across market participants) of $19,000 per event for SPY and roughly $50,000 per event for ES. This translates to roughly $15 million in cumulative profits on average each year, which is trivial relative to about $4.7 trillion traded in SPY and $35.8 trillion notional value traded in ES in 2012. The $15 million is also trivial compared with the cost of price discovery in U.S. markets, which at 0.67% of the market capitalization (French 2008) amounted to roughly $100 billion in 2006.").

1692 See Proposing Release, 85 FR at Section IV.A. The hub-and-spoke model of the exclusive SIPs exacerbates this geographic latency. See supra note 676. See also MEMX Letter at 6.

1693 See Proposing Release, 85 FR at n.396.

Aggregation or consolidation latency refers to the amount of time an exclusive SIP takes to aggregate the multiple sources of SRO market data into SIP data and includes the time it takes to calculate the NBBO. This latency reflects the time interval between when an exclusive SIP receives data from an SRO and when it disseminates consolidated data to the end-user. Even though in recent years the exclusive SIPs made improvements to address aggregation latency, the proposal stated that the related latency differential remains; as mentioned above, in the second quarter of 2019, for Tapes A and B average quote feed and average trade feed aggregation latencies were 69 and 139 microseconds, respectively.[1694] In the same time period, the Tape C aggregation latency was an average of 16.9 microseconds for quotes and 17.5 microseconds for trades.[1695] Notably, these latency differentials remain even though the Equity Data Plans' Operating Committees have made some improvements to certain aspects of the exclusive SIPs and related infrastructure, including improvements to address aggregation latency.[1696]

One commenter pointed out that the CTA SIP has implemented improvements to its processing, which at the time that the commenter expected to bring the aggregation time down to "under 20 microseconds."[1697] While these improvements will likely reduce the aggregation latency of the CTA SIP, 20 microseconds of aggregation latency will continue to be meaningfully slower than current market practice in the aggregation of proprietary data feeds, and only about as fast as the UTP SIP is currently.

Transmission latency refers to the time interval between when data is sent (e.g., from an exchange) and when it is received (e.g., at an exclusive SIP and/or at the data center of the

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1694 See Proposing Release, 85 FR at Section IV.A.

1695 Id.

1696 See Proposing Release, 85 FR at Section IV.A.

1697 See NYSE Letter II at 10, 11.

subscriber), and the transmission latency between two fixed points is determined by the transmission communications technology through which the data is conveyed. Transmission latency also varies depending on the geographic distance between where the data is sent and where it is received. There are several options currently used for transmitting market data, such as fiber optics, which typically are used by the exclusive SIPs for receipt and dissemination of SIP data, and wireless microwave connections, which the exchanges offer as an alternative for their proprietary data feeds but not for SIP data.[1698] Fiber optics are generally more reliable than wireless networks since the data signal is less affected by weather. The modes of transmission for SIP data are typically slower than the modes of transmission used for proprietary data. In the Proposing Release, the Commission stated that each of the CTA/CQ Plan participants must transmit its data through connectivity options that have a round-trip latency of at least 280 microseconds.[1699] One of the commenters said that "[i]n 2019, the SIP Operating Committee authorized two improvements to the CTA SIP" and that this change "will reduce what the Commission refers to as CTA SIP data 'transmission' latency, i.e., the time interval between when the data is sent and when it is received, by over 140 microseconds."[1700]

The Commission believes that the benefits of greater speed on the timescales at which the market currently measures latency have mostly to do with being faster than one's competitors. In some situations small latency differentials that leave enough time for certain market participants to observe and react to information before other, slower market participants can be

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1698 Id.

1699 See Proposing Release, 85 FR at n.410.

1700 NYSE Letter II at 10–11.

as costly to slower market participants as larger latency differentials.[1701] For example, a market participant may use market data to anticipate price movements and then place limit orders ahead of the price movement. In doing so, the market participant will end up in a queue of limit orders placed in the book, in order of time priority. If other market participants react the same way, then this market participant's quote will be behind the quotes in the queue of those who reacted faster. If the market participant increases its reaction time but still does not end up faster than the trader who placed the order directly in front of it in the queue, then the market participant's quote will have the exact same priority that it had at the slower reaction time.

Currently, some market participants obtain proprietary data feeds from many SROs.[1702] Of these market participants, some prefer to have consolidated proprietary data. There are two ways these market participants can obtain consolidated data. First, market participants may

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1701 Academic literature examines the effects of trading speed on revenues, adverse selection, and liquidity. See, e.g., Matthew Baron et al., *Risk and Return in High-Frequency Trading*, 54 J. FIN. & QUANTITATIVE ANALYSIS 993 (2019) (testing the connection between high frequency trading ("HFT") latency and trading performance; the authors find that relative latency matters and that "HFT firms exhibit large, persistent cross-sectional differences in performance, with trading revenues disproportionally accumulating to a few firms." Furthermore, when HFT firms use their relative latency advantages to trade on news to create short-term arbitrage opportunities, they generate adverse selection on slower traders.); Bruno Biais et al., *Equilibrium fast trading*, 116 J. FIN. ECON. 292 (2015) (arguing that fast trading technology "provides advance access to value-relevant information, which creates adverse selection, lowering welfare," and "generates a negative externality"); Thierry Foucault et al., *Toxic Arbitrage*, 30 REV. FIN. STUD. 1053 (2017) (providing evidence that "[a]rbitrage opportunities due to asynchronicities in the adjustment of prices to news are toxic because they expose dealers to the risk of trading with arbitrageurs at stale quotes." The authors then claim that these toxic arbitrage opportunities that come with higher trading speed impair market liquidity.).

1702 The exchanges, as a subset of SROs, sell proprietary data feeds to market participants.

independently create consolidated data by purchasing individual exchange proprietary market data products and consolidating that information for their own use.

Second, market participants may obtain consolidated data from market data aggregators, which are mostly firms that purchase direct access to exchange data,[1703] consolidate the data, and disseminate the data (after various levels of processing) to market participants.[1704] Additionally, some market data aggregators do not purchase direct access to exchanges. Instead, they provide hardware and software for market data aggregation to the parties that have contractual relationships to purchase or license the market data enabling market participants to outsource the significant hardware, software, and personnel expertise that is required to consolidate the proprietary feeds directly. Many of the most sophisticated market participants in the market use these products, and despite the fact that they create an additional chain link between market participants and proprietary feeds, the Commission believes that these firms generally deliver data to market participants faster than the exclusive SIPs.[1705]

Market participants who subscribe to SIP data also have two different ways of obtaining their data. They can either directly get SIP data feeds from the exclusive SIPs or, as stated by a

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1703 As mentioned below, even when obtaining consolidated market data from market data aggregators, market participants also have to pay data fees directly to the exchanges. See infra Section V.B.2(c).

1704 Market participants who consolidate market data independently may use other market data aggregators' products and services such as software.

1705 See, e.g., Roundtable Day One Transcript at 128–29 (Mark Skalabrin, Redline Trading Solutions).

commenter,[1706] they can get SIP data from a third party aggregator in a normalized form.[1707] The least latency sensitive market participants are the most likely to receive SIP data in this normalized form.

Some commenters stated that the need for backup data feeds is an important cost in obtaining access to market data.[1708] The Commission believes that today, many market participants use the exclusive SIPs as a backup, and maintain a subscription to the exclusive SIP feeds despite using proprietary data for trading decisions partly for this reason. The exclusive SIPs themselves maintain a geographically diverse backup system consistent with Regulation SCI. One participant in the Market Data Roundtable stated that the exclusive SIPs are "expensive for a backup feed."[1709]

(c) Current State of Utilization of Market Data

One commenter stated that the introduction of different levels of quality in core data consolidation and dissemination would introduce new "tiers" into the market beyond the two tiers of those who use proprietary DOB feeds and those who do not.[1710] The Commission does not believe that differing tiers of market data access sophistication and technology represent changes to the market, given current market practice. Market participants have different levels of sophistication in receiving and processing real time market data, because of differences in the

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1706 BestEx Research Letter at 8.

1707 Companies that normalize market data take in raw data delivered in a variety of protocols and, using feed handlers, normalize it into a single protocol different from the one used by the original venue. This way a data user can receive one feed using one streaming protocol. See Vela's *Definitive Guide to Market Data*, available at https://info.tradevela.com/definitive-guide-to-market-data#normalisation.

1708 See, e.g., Angel Letter at 20; NYSE Letter II at 24; FINRA Letter at 4.

1709 Roundtable Day One Transcript at 140 (Mark Skalabrin, Redline Trading Solutions).

1710 See, e.g., Nasdaq Letter IV at 8.

cost of maintaining data processing systems and in the data needs of various trading and investment strategies. More sophisticated firms use advanced data access methods and technologies, and generally seek to reduce latency and improve the way in which the data can be used. Other market participants trade latency for lower costs, and this has resulted in a continuum of different levels of latency and processing quality in the market.

The most competitive executing broker-dealers, market makers, and traders using highly latency sensitive strategies define market practice at the highest-cost, lowest-latency end of the continuum. These market participants typically invest significantly more resources in reducing latency and increasing processing speed than any other kind of market participant. This group typically purchases co-location services at all major data centers, along with the highest capacity connectivity services and the most raw and unprocessed exchange proprietary data feeds.[1711] Many market participants in this group maintain their competitive advantage by performing all major steps related to data connection and processing within their own business. That is, they arrange connectivity, software, hardware, and transmission necessary to receive and process market data on their own without employing the services of outside vendors. As a result, there are highly significant technological, infrastructure, and personnel costs to building and

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[1711] For example, NYSE describes their order-by-order message feed, NYSE Integrate Feed, as a "high-performance product." See https://www.nyse.com/market-data/real-time/integrated-feed (last accessed Sept. 21, 2020).

maintaining such a system for data processing.[1712] Therefore, the Commission believes that there are relatively few market data users at this level.[1713]

Market participants that seek to reduce the costs of maintaining this high level of capability in market data access make a variety of cost saving adjustments. For example, market participants may decide to employ vendors to assist in the most difficult or sophisticated aspects of the process, such as microwave transmission and hardware. These market participants may also use software vendors to aggregate proprietary data, and may also employ vendors to assist in connecting to the data feeds. While using such vendors can reduce cost, this can sometimes come at the expense of adding latency.[1714] This can happen because market participants may base their competitive advantage on the development of technology, which might be superior to what is available from vendors, or because the level of customizability and specialization to the

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1712 As an example of such costs, see *What Types of Financial Market Data Providers Are There?*, EXEGY BLOG, available at https://www.exegy.com/2019/07/types-financial-market-data-providers/ (last accessed Sept. 21, 2020), stating that "[e]xchanges are the most expensive provider option," because data directly from the exchange comes raw and in whatever format the exchange uses, and this leaves the end user of the data with the task of "maintaining, transporting and processing the data." Because firms that are the most sophisticated users of market data consume data directly from the exchange, these are costs they incur.

1713 The market participants at this level are a subset of all market participants using proprietary data (see supra Section V.B.2(a)). Much of the text discusses market participants who use proprietary data feeds; the different levels on the continuum consist of differences in how those feeds are used even within the set of market participants who use proprietary data.

1714 See Vela, supra note 1707, stating that the use of "…normalized feed of data from a vendor, however, can add latency, which may make it less suitable for latency-sensitive applications…"